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Synthetic Sensor Data Generation Framework

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Software

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Summary

Field datasets rarely include ground-truth labels for sensor errors, making it difficult to systematically develop and validate methods for anomaly detection, data quality control, and sensor diagnostics.

This package provides an **R Shiny–based framework** for generating **synthetic time series from two correlated or redundant sensors**, with **user-configurable background signals** and **injected anomalies** (spikes, drifts, delay/attenuation, noise, and bias). The output includes both **anomaly labels** and **true error magnitudes**, enabling quantitative evaluation of diagnostic algorithms.

The framework supports both interactive use through a Shiny app and programmatic use through R functions, making it useful for researchers and practitioners in sensor-driven domains. Its main goal is to facilitate **reproducible benchmarking** of sensor data quality assessment techniques in situations where controlled evaluation datasets are otherwise scarce.

Statement of need

While real-world sensor datasets are valuable, they rarely include complete ground-truth labels for error types and magnitudes. This limits the ability to systematically evaluate diagnostic algorithms.

This software addresses these gaps by:

- Allowing **two sensors** to measure either the same or correlated background signals.
- Providing a **broad set of configurable anomaly types**: spikes, drifts, sensor delay/attenuation, noise, and constant bias.
- Enabling **precise control over anomaly magnitude, duration, and correlation** between sensors.
- Outputting **CSV files with labeled anomalies** for use in supervised learning and method validation.
- Offering an **interactive Shiny UI** for rapid experimentation, as well as scriptable functions for batch generation.

Related work

Several tools provide synthetic time-series data or anomaly-related functionality, but to our knowledge none combine **multi-sensor redundancy**, **configurable anomaly injection**, and **ground-truth error magnitudes**. Table 1 summarizes the most relevant approaches.

Tool / Package	Key Features	Limitations vs. our framework
SimTensor (MATLAB) (Fanaee-T & Gama, 2016)	Generates synthetic tensors with periodicity, seasonal patterns, and change-points.	No sensor redundancy; no labeled error magnitudes; not designed for anomaly benchmarking.
GAN-based generators (e.g. TimeGAN, Python) (Yoon et al., 2019)	Produces realistic synthetic time series via deep generative models.	Relies on training data. No explicit anomaly type, magnitude, or timing control; no labeled errors.
Mutation-based anomaly frameworks (e.g. MDPI 2024) (Jeon et al., 2024)	Mutates multivariate series to introduce rare events.	Limited anomaly control; no error magnitudes; no multi-sensor correlation.
R packages for anomaly detection (`anomalize`, `otsad`) (Dancho & Vaughan, 2023, p. otsad2019)	Detect anomalies in real or synthetic data.	Focus on detection, not generation; no ground truth labels or anomaly injection.
MATLAB/Simulink examples (MathWorks, 2025)	Demonstrate anomaly detection in streaming multichannel signals.	Simple signals; no flexible anomaly generation; no dataset export.

Table 1: Comparison of related tools with respect to anomaly generation and benchmarking functionality.

Use

The framework has been adopted in several research projects. It is used in (Skålvik et al., n.d.) to generate controlled datasets for validating a Bayesian method for error and uncertainty estimation of environmental sensor data. In (Heggedal, 2025), it was applied in the development of a software tool that assists experts with annotating marine time series. In (Stuen et al., 2025) it was used for demonstrating a method for uncertainty quantification of hydrogen leaks.

Beyond these projects, the package is broadly useful for:

- **Reproducible benchmarking**: researchers can share identical datasets (using fixed seeds) to compare diagnostic methods.
- **Teaching and training**: students can experiment with sensor error effects without requiring access to field instruments.
- **Rapid prototyping**: practitioners can quickly simulate "what if" scenarios for sensor anomalies before deploying to the field.

Implementation

The framework is implemented in **R** and is provided both as:

1. A **Shiny application** for interactive configuration and dataset export.
2. A set of **modular R functions** for programmatic dataset generation.

Background signal

The package provides functions to generate several background signal models:

- **Autoregressive process (AR(1))** — for stationary signals with tunable autocorrelation.
- **Random walk** — a special case of AR(1) with strong temporal autocorrelation.
- **Poisson moving average** — for smoothed stochastic signals.
- **Sine wave** — for periodic variation.

For the AR(1) and Random walk, the background signals can be **correlated** between sensors with user-defined cross-correlation. For the Poisson moving average and the Sine wave, the background signals are identical between the sensors.

Anomaly types

Anomalies can be introduced individually or in combination:

- **Delay/attenuation** — models exponential moving average sensor response.
- **Noise and constant bias** — configurable per sensor, with optional correlation between noise terms.
- **Spikes** — short-duration deviations, correlated or uncorrelated between sensors, with random sign and magnitude.
- **Drifts** — gradual deviations over time, with configurable slope and duration.

Anomalies are inserted at random positions while ensuring spacing from time-series edges to avoid boundary artifacts.

Output

The output is a **CSV file** containing:

- Time series for both sensors.
- Flags for each time step indicating anomaly type.
- True error magnitude for each sensor.

Reproducibility

All random processes are controlled via a user-specified random seed. Full mathematical details of signal generation and anomaly injection are provided in the software documentation and supplementary materials (https://github.com/AstridMarie2/synthsensor –> *Synthetic Sensor Data Generation Framework - Mathematics*).

All functionality is covered by automated tests (testthat) and continuous integration (GitHub Actions).

Example

An example workflow for generating a dataset with correlated AR(1) background signal, additive Gaussian noise, and occasional spikes:

```
library(synthsensor)

df <- generate_data_function(
  n = 300,
```

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```r
  add_background   = TRUE,
  background_type  = "Sine Wave",
  sine_amplitude   = 2,
  sine_period      = 30,
  sd1 = 0.15, sd2 = 0.15,
  crosscor_noise   = 0,
  mean1 = 0.5, mean2 = -0.5,
  # anomalies
  n_spikes_corr    = 0,
  n_spikes_s1      = 3,
  n_spikes_s2      = 2,
  max_spike_length = 6,
  n_drifts_s1      = 1,
  n_drifts_s2      = 0,
  drift_duration   = c(25, 35),
  drift_slope      = c(0.05, 0.08),
  delayed_sensor   = "None",
  alpha_ema        = 0.3
)

head(df)

#>   TimeSinceClean  Sensor1    Sensor2               Date AnomalyFlag1 AnomalyFlag2 Me
#> 1              1 1.024476 -0.1734142 2025-01-01 00:00:00       Normal       Normal  0
#> 2              2 1.427743  0.2737805 2025-01-01 01:00:00       Normal       Normal  1
#> 3              3 1.817718  0.9042105 2025-01-01 02:00:00       Normal       Normal  1
#> 4              4 2.145533  0.9916997 2025-01-01 03:00:00       Normal       Normal  1
#> 5              5 2.298991  1.2462778 2025-01-01 04:00:00       Normal       Normal  2
#> 6              6 2.353803  1.6542066 2025-01-01 05:00:00       Normal       Normal  2


write.csv(df, "synthetic_dataset.csv", row.names = FALSE)
```

Alternatively, launch the Shiny app:

```r
synthsensor::app_synth()
```

Dependencies

- R (>= 4.0)
- shiny
- dplyr
- ggplot2
- MASS
- stats
- gridExtra
- zoo

Availability

The framework is available as:

1. An online Shiny app: https://sensordiagnostics.shinyapps.io/app_synth/

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2. Source code and documentation on GitHub: https://github.com/AstridMarie2/synth-sensor

Zenodo DOI link: https://doi.org/10.5281/zenodo.17157665

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References

Dancho, M., & Vaughan, D. (2023). Anomalize: Tidy anomaly detection. *R Package Version 0.3.0*. https://doi.org/https://doi.org/10.32614/CRAN.package.anomalize

Fanaee-T, H., & Gama, J. (2016). *SimTensor: A synthetic tensor data generator*. https://arxiv.org/abs/1612.03772

Heggedal, S. B. (2025). *A software tool-assisted approach to event detection and annotation in marine time series* [Master's Thesis]. Western Norway University of Applied Sciences.

Jeon, S., Koo, K., Moon, D., & Seo, J. T. (2024). Mutation-based multivariate time-series anomaly generation on latent space with an attention-based variational recurrent neural network for robust anomaly detection in an industrial control system. *Applied Sciences*, *14*(17). https://doi.org/10.3390/app14177714

MathWorks. (2025). *Deep signal anomaly detector*. https://www.mathworks.com/help/dsp/ref/deepsignalanomalydetector.html.

Skålvik, A. M., Bjørk, R. N., Parvilainen, P., Frøysa, K.-E., & Saetre, C. (n.d.). *A bayesian approach for error estimation and uncertainty assessment of oceanographic sensor data*. Manuscript in preparation.

Stuen, I. M., Sætre, C., & Folgerø, K. (2025). *Measurement uncertainty and time correlation effects on mass balance thresholds for detecting leaks in hydrogen networks*. Mathmet 2025, Gothenburg, Sweden.

Yoon, J., Jarrett, D., & Schaar, M. van der. (2019). Time-series generative adversarial networks. In H. Wallach, H. Larochelle, A. Beygelzimer, F. dAlché-Buc, E. Fox, & R. Garnett (Eds.), *Advances in neural information processing systems* (Vol. 32). Curran Associates, Inc. https://proceedings.neurips.cc/paper_files/paper/2019/file/c9efe5f26cd17ba6216bbe2a7d26d490-Paper.pdf

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